

Evaluating Network Topology Tradeoffs in Gem5 Using Built-In and Custom Interconnects

SooHyuk Cho
Princeton University
Princeton, NJ, USA
soohyuk.cho@princeton.edu

Daniel Li
Princeton University
Princeton, NJ, USA
dl4247@princeton.edu

Abstract—
Index Terms—

I. PROJECT OVERVIEW

We propose to study how different on-chip network topologies affect communication latency, throughput, and scalability in small multiprocessor systems using the gem5 simulator, a widely used (micro)architectural simulator [1]. Rather than building a new network model from scratch, we will reuse gem5's existing Ruby/Garnet interconnection-network infrastructure [2] and extend it with a small set of custom topologies. Our goal is to compare both standard and custom-designed topologies under controlled synthetic traffic in order to understand the tradeoffs between metrics such as network diameter, path diversity, concentration, and congestion behavior [3].

The project will use gem5's Garnet synthetic-traffic framework as the main evaluation environment. This allows us to focus on the interconnect itself without taking on the additional complexity of a full cache-coherent multicore software stack. We will begin by reusing several built-in topologies already supported by gem5, including a crossbar-based baseline and mesh-based baselines, and then implement custom topology files for a *ring* and a *tree* topology. If time permits, we will also implement a *star* topology as a third custom design. This gives the project a clear custom contribution while keeping the scope manageable and centered on topology design and evaluation rather than a full simulator re-engineering.

II. PROJECT PLAN

The main objectives of this project are:

- Compare built-in and custom network topologies using a common gem5 Garnet evaluation framework.
- Quantify how topology choice affects average packet latency, throughput, and saturation behavior.
- Study how topology tradeoffs change as system size increases.
- Reuse gem5's existing routing and topology infrastructure as much as possible, adding only lightweight custom topology definitions instead of redesigning the network stack.
- Provide one clear custom contribution by implementing at least two new topologies in gem5.

A. Topologies to Compare

We plan to compare the following topologies:

- **CrossbarGarnet**: a low-diameter built-in baseline with high connectivity [2].
- **Mesh_XY**: a standard scalable 2D mesh baseline.
- **Mesh_westfirst**: a mesh with different routing behavior, useful for separating routing effects from pure topology effects.
- **Ring**: custom topology with low implementation complexity and higher diameter.
- **Tree**: custom topology that offers a different hierarchical communication structure.
- **Star**: optional third custom topology if schedule permits.

Our primary custom contribution will be the implementation of the **Ring** and **Tree** topologies. The **Star** topology will be included if development remains on schedule. If time becomes limited, Ring and Tree will be prioritized over Star.

B. Implementation Plan

To keep the project feasible, we will reuse gem5's existing Garnet standalone infrastructure and avoid unnecessary modifications to the simulator core.

- We will use the existing traffic configuration flow (e.g. `garnet_synth_traffic.py`) as the main driver for experiments.
- We will implement custom topologies as new gem5 topology files, following the style of the existing built-in topology definitions.
- We will initially rely on gem5's default weight-based routing-table mechanism rather than implementing a new routing algorithm.
- If time permits, we may add one small routing-related extension, but this is not part of the core project scope as of now.

This design choice keeps the project focused on architectural comparison rather than low-level simulator debugging.

C. Experimental Methodology

We will evaluate each topology under a common experimental framework. The main factors we will vary are:

- **Node counts**: 8, 16, and 32 nodes

- **Traffic patterns:** `uniform_random`, `neighbor`, `transpose`, and `tornado`
- **Injection rate:** swept from low load up to the onset of congestion

To add more architectural depth without making the project too large, we will also include one small sensitivity study. Our current preference is to vary one of the following:

- router latency, or
- number of virtual channels per virtual network.

All other simulation settings will be held constant when making a comparison so that observed differences can be attributed to topology rather than unrelated configuration changes.

D. Metrics

We plan to report the following metrics (but not limited to):

- Average packet latency
- Throughput as offered load increases
- Approximate saturation point for each topology
- qualitative fairness and congestion behavior under asymmetric traffic

We will also discuss the likely implementation tradeoffs of each topology, such as connectivity, likely router degree, and scalability trends, even when those costs are not directly modeled as area or energy.

E. Expected Contributions

We expect the project to make the following contributions:

- A reusable `gem5`-based evaluation framework for comparing network topologies under synthetic traffic
- A comparative study of several built-in and custom topologies
- New `gem5` topology implementations for Ring and Tree, with Star as an optional extension
- A clear analysis of which topologies are most effective under different communication patterns and scales

REFERENCES

- [1] N. Binkert, B. Beckmann, G. Black, S. K. Reinhardt, A. Saidi, A. Basu, J. Hestness, D. R. Hower, T. Krishna, S. Sardashti, R. Sen, K. Sewell, M. Shoaib, N. Vaish, M. D. Hill, and D. A. Wood, "The `gem5` simulator," *SIGARCH Comput. Archit. News*, vol. 39, p. 1–7, Aug. 2011.
- [2] N. Agarwal, T. Krishna, L.-S. Peh, and N. K. Jha, "Garnet: A detailed on-chip network model inside a full-system simulator," in *2009 IEEE International Symposium on Performance Analysis of Systems and Software*, pp. 33–42, 2009.
- [3] L. Benini and G. De Micheli, "Networks on chips: A new soc paradigm," *Computer*, vol. 35, p. 70–78, Jan. 2002.